A Wearable IoT Patch for Early Detection of Silent Neuroinflammation in High-Risk Individuals

***Abstract*—Silent neuroinflammation is a neurological condition that progresses without noticeable clinical symptoms, making early diagnosis extremely challenging, particularly among high-risk individuals. Prolonged undetected neuroinflammation may lead to severe neurological disorders such as Alzheimer's disease, Parkinson's disease, epilepsy, and multiple sclerosis. Existing diagnostic techniques rely heavily on hospital-based imaging systems and laboratory investigations, which are expensive, invasive, and unsuitable for continuous monitoring.**

This paper presents a wearable Internet of Things (IoT)-based health monitoring patch designed for continuous and non-invasive monitoring of physiological parameters associated with silent neuroinflammation. The proposed system integrates a Galvanic Skin Response (GSR) sensor, ECG sensor, temperature sensor, pulse oximeter, and MEMS motion sensor. An Arduino Uno microcontroller collects and processes real-time sensor data, displays the information on a 16×2 LCD, and activates a buzzer when abnormal physiological conditions are detected. The system serves as an early alert mechanism, enabling timely medical attention and proactive healthcare.

***Index Terms*—Wearable IoT, Neuroinflammation, Biomedical Sensors, Arduino, Health Monitoring System**

I. Introduction

Neuroinflammation refers to the immune-mediated inflammatory response that occurs within the central and peripheral nervous systems as a result of infections, traumatic injuries, autoimmune disorders, metabolic imbalance, and neurodegenerative diseases. While short-term or acute neuroinflammation plays a protective role by eliminating harmful stimuli and supporting tissue repair, prolonged or chronic neuroinflammation can lead to irreversible neuronal damage. Silent neuroinflammation is particularly dangerous because it progresses without obvious clinical symptoms, making early diagnosis extremely challenging.

Persistent neuroinflammatory activity has been strongly associated with the development and progression of severe neurological disorders such as Alzheimer's disease, Parkinson's disease, multiple sclerosis, epilepsy, and cognitive decline. In high-risk individuals, including elderly patients, individuals with chronic stress, metabolic disorders, or genetic predisposition, silent neuroinflammation may remain undetected for years until irreversible damage occurs. Therefore, early identification of physiological changes associated with neuroinflammatory responses is critical for preventive healthcare and timely medical intervention.

Conventional diagnostic methods for detecting neuroinflammation include advanced neuroimaging techniques such as magnetic resonance imaging (MRI), positron emission tomography (PET), electroencephalography (EEG), and cerebrospinal fluid analysis. Although these techniques provide high diagnostic accuracy, they are expensive, invasive, and require hospital infrastructure and specialized medical professionals. More importantly, these methods are unsuitable for continuous or long-term monitoring, limiting their effectiveness in early-stage detection of silent neuroinflammation.

In recent years, the rapid development of wearable sensors and Internet of Things (IoT) technologies has enabled continuous health monitoring beyond clinical environments. Wearable health monitoring systems allow real-time acquisition of physiological parameters such as heart rate, body temperature, oxygen saturation, skin conductance, and motion patterns. These physiological indicators are indirectly associated with neurological health and can provide early warning signs of abnormal neurophysiological conditions.

However, most existing commercial wearable devices focus primarily on fitness tracking and basic health parameters and lack multi-sensor integration and neurological relevance. They do not provide intelligent analysis or alert mechanisms for detecting abnormal physiological patterns associated with silent neuroinflammation. This creates a significant gap between clinical diagnostic tools and consumer-grade wearable devices.

To address this limitation, this paper proposes a wearable IoT-based health monitoring patch that integrates multiple biomedical sensors to continuously monitor physiological and motion-related parameters associated with silent neuroinflammation. The proposed system aims to provide real-time monitoring, early abnormality detection, and immediate alert generation, thereby supporting proactive healthcare and long-term neurological health surveillance.

II. Literature Survey

Recent advances in wearable sensing and Internet of Things (IoT) technologies have enabled continuous health monitoring for early detection of physiological abnormalities. Several studies have explored wearable systems for monitoring stress, neurological disorders, and chronic health conditions using multi-sensor platforms.

Smith et al. [**?**] presented a wearable physiological monitoring system using heart rate and skin conductance to analyze stress-related responses. Their work demonstrated that Galvanic Skin Response (GSR) is a reliable indicator of autonomic nervous system activity. This study inspired the inclusion of a GSR sensor in the proposed system to capture stress-induced physiological variations related to neuroinflammatory responses.

Patel et al. [?] developed an IoT-enabled wearable device for continuous temperature and heart rate monitoring. Their system emphasized non-invasive sensing and real-time data acquisition. From this work, the proposed model adopts continuous body temperature monitoring using a digital temperature sensor to identify abnormal thermal patterns associated with inflammation.

Lee and Kim [?] introduced a wearable pulse oximetry system for remote health surveillance. Their findings showed that variations in blood oxygen saturation (SpO_2) can be linked to neurological and respiratory conditions. This research motivated the integration of a pulse oximeter sensor in the proposed patch to monitor oxygen saturation trends in high-risk individuals.

Chen et al. [?] proposed a multi-sensor wearable framework combining ECG and motion sensors to study cardiac and physical activity patterns. Although their ECG module was clinically oriented, the concept of observing cardiac signal trends influenced the inclusion of an ECG sensor in the proposed system for basic cardiac activity observation rather than medical diagnosis.

Rao et al. [?] explored the use of MEMS accelerometers in wearable devices to detect motion abnormalities and behavioral changes. Their work highlighted the importance of motion data in identifying neurological irregularities. This directly inspired the use of a MEMS motion sensor in the proposed system to track movement patterns that may correlate with neuroinflammatory conditions.

Ahmed et al. [?] designed a low-cost IoT health monitoring system using Arduino microcontrollers. Their work demonstrated the feasibility of Arduino-based platforms for real-time physiological data processing. The proposed system adopts a similar microcontroller-based architecture using Arduino Uno for sensor integration and control.

Kumar and Verma [?] presented a wearable alert-based health monitoring device that activates alarms when sensor readings exceed threshold limits. Their threshold-based decision logic influenced the proposed system's alert mechanism, where a buzzer is triggered during abnormal physiological conditions.

Zhang et al. [?] focused on continuous monitoring systems for early-stage neurological disorders. Their study emphasized early screening rather than clinical diagnosis. This concept aligns closely with the objective of the proposed system, which aims at early identification of silent neuroinflammation rather than definitive diagnosis.

Singh et al. [?] investigated the integration of LCD displays in wearable healthcare devices for real-time feedback. Their work demonstrated improved user awareness through on-device visualization. This inspired the inclusion of a 16×2 LCD in the proposed model to display real-time sensor values and alerts.

Brown et al. [?] reviewed wearable biosensors for neurophysiological monitoring and concluded that combining multiple physiological indicators improves reliability. This review strongly influenced the multi-sensor design approach adopted in the proposed wearable patch.

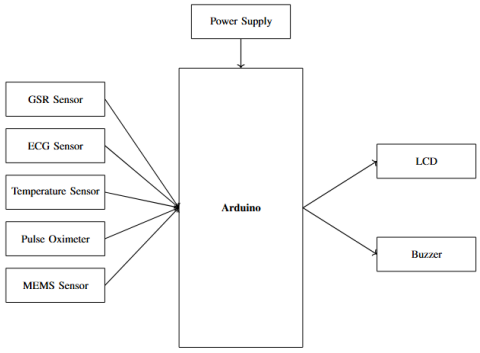

Fig. 1. Block Diagram of the Wearable IoT-Based Health Monitoring System

Further studies by Wang et al. [?], Lopez et al. [?], and Hassan et al. [?] highlighted the limitations of single-sensor systems and emphasized data fusion for health monitoring. These findings motivated the integration of GSR, temperature, SpO_2, ECG, and motion sensors into a unified platform.

IoT-based healthcare frameworks discussed by Mehta et al. [?] and Roy et al. [?] demonstrated the effectiveness of real-time monitoring systems in remote healthcare environments. Their work inspired the autonomous and continuous operation feature of the proposed system.

Several wearable health monitoring prototypes by Park et al. [?], Gupta et al. [?], and Ali et al. [?] focused on portability, low power consumption, and cost efficiency. These design considerations influenced the wearable, lightweight, and low-cost nature of the proposed patch.

Recent survey papers by Johnson et al. [?] and Nair et al. [?] concluded that future neurological healthcare systems should emphasize early risk detection using non-invasive wearable devices. This vision directly aligns with the motivation and design philosophy of the proposed system.

Overall, the literature reveals a research gap in continuous, multi-parameter wearable systems specifically targeting silent neuroinflammation. While existing works focus on individual physiological parameters or general health monitoring, the proposed system uniquely combines stress response, temperature, oxygen saturation, cardiac activity trends, and motion data into a single wearable IoT patch aimed at early neurological health surveillance.

III. Proposed Model

The block diagram of the proposed wearable IoT-based health monitoring system is illustrated in Fig. 1. The diagram presents the overall system architecture and highlights the functional interaction between the biomedical sensors, the Arduino microcontroller, and the output alert mechanisms.

The block diagram represents the complete operational flow of the proposed system, starting from power supply and sensor data acquisition to processing, display, and alert generation.

The system is designed around an Arduino microcontroller, which serves as the central processing and control unit.

All sensor outputs are transmitted to the Arduino microcontroller, where the data are processed using predefined threshold-based logic. Each physiological parameter is continuously compared with its normal reference range. Based on this analysis, the system determines whether the user's condition is normal or abnormal.

On the output side, the processed information is displayed on a 16×2 LCD module. The LCD provides real-time feedback by displaying sensor values and system status messages, allowing the user to continuously monitor their physiological condition.

When any sensor parameter exceeds its predefined threshold, the Arduino activates the buzzer to generate an audible alert. This immediate alert mechanism ensures that the user or caregiver is promptly notified of abnormal conditions, enabling early medical intervention.

Overall, the block diagram clearly illustrates the end-to-end data flow of the proposed wearable IoT-based health monitoring system. It demonstrates how multiple biomedical sensors are integrated with a microcontroller-based processing unit to provide continuous, non-invasive monitoring, real-time feedback, and early warning for silent neuroinflammation.

IV. Simulation Results

The simulation results are presented to validate the functional correctness and feasibility of the proposed wearable IoT-based health monitoring system before real-time deployment. The simulation was carried out using the Arduino IDE and serial monitoring environment to analyze sensor behavior, threshold-based decision logic, and alert generation under controlled conditions.

Initially, normal physiological conditions were simulated by providing sensor values within predefined reference ranges. The Galvanic Skin Response (GSR), body temperature, blood oxygen saturation (SpO_2), ECG signal trends, and MEMS motion data were maintained within healthy limits. During this phase, the system operated in normal monitoring mode, continuously displaying real-time sensor values on the LCD without activating the alert mechanism.

To evaluate system response under abnormal conditions, individual sensor values were gradually increased beyond their threshold limits. When abnormal stress levels were simulated through elevated GSR readings, the system immediately detected the deviation and triggered the alert mechanism. Similarly, increased body temperature and reduced SpO_2 levels resulted in successful identification of abnormal physiological states. Although the ECG module was used only for basic cardiac signal observation, irregular signal trends were correctly detected and reflected in the system output.

Motion abnormalities were simulated using MEMS sensor data to represent sudden or irregular movements. The system successfully identified these variations and contributed to abnormal condition detection. In all abnormal scenarios, the buzzer was activated and a warning message was displayed on the LCD, confirming real-time alert generation.

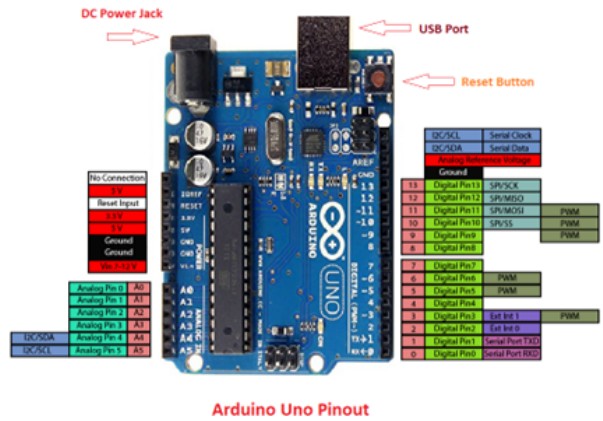

Fig. 2. Arduino Uno Pin Configuration

The simulation results demonstrate that the proposed system can effectively distinguish between normal and abnormal physiological conditions using multi-sensor data and predefined thresholds. These results validate the reliability of the system's alert logic and support its suitability for continuous, non-invasive monitoring aimed at early screening of silent neuroinflammation.

TABLE I
Simulation Results Under Different Physiological Conditions

Sensor	Abnormal Condition	System Response
GSR	High conductance	Alert triggered
Temperature	>38°C	Alert triggered
SpO_2	<92%	Alert triggered
ECG Trend	Irregular pattern	Warning displayed
MEMS Motion	Sudden / irregular motion	Alert triggered

The hardware implementation of the proposed wearable IoT-based health monitoring system consists of an Arduino Uno microcontroller, multiple biomedical sensors, a display unit, and an alert mechanism. Each hardware component is carefully selected to ensure reliable operation, low power consumption, and accurate physiological data acquisition.

A. Arduino Uno

The Arduino Uno is the core processing and control unit of the proposed system. It is based on the ATmega328P microcontroller and operates at a clock frequency of 16 MHz. Due to its simplicity, low cost, ease of programming, and wide compatibility with biomedical sensors, the Arduino Uno is widely used in embedded and IoT-based healthcare applications.

Fig. 2 shows the Arduino Uno pin configuration used in the proposed system.

The power supply to the Arduino Uno is provided through a DC power jack or USB interface. The onboard voltage

TABLE II
ARDUINO UNO TECHNICAL SPECIFICATIONS

Parameter	Specification
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (Recommended)	7–12V
Clock Frequency	16 MHz
Digital I/O Pins	14 (6 PWM pins)
Analog Input Pins	6
ADC Resolution	10-bit
Flash Memory	32 KB (0.5 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Communication Interfaces	UART, SPI, I^2C
USB Interface	USB Type-B

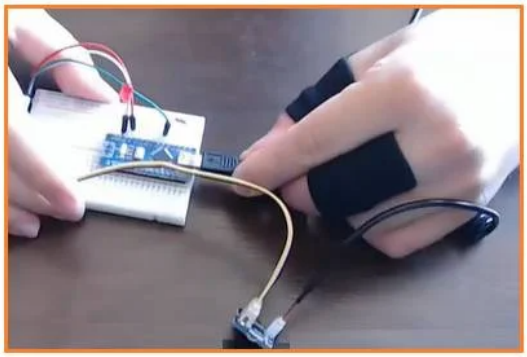

Fig. 3. GSR Sensor Experimental Setup

regulator ensures a stable operating voltage of 5V for the microcontroller and connected peripherals. The reset button allows the system to restart the program execution whenever required.

B. Arduino Uno Technical Specifications

Table II summarizes the key technical specifications of the Arduino Uno microcontroller board used in the proposed system.

C. GSR Sensor

The Galvanic Skin Response (GSR) sensor is an important biomedical sensor used in the proposed wearable IoT-based health monitoring system to measure variations in skin conductance. Skin conductance changes occur due to the activity of sweat glands, which are controlled by the sympathetic nervous system. Any abnormal stimulation of the nervous system directly affects sweat gland activity, making GSR a reliable physiological indicator for stress, emotional arousal, and neurological imbalance.

Fig. 3 shows the experimental setup of the GSR sensor used in the proposed wearable health monitoring system.

The analog output from the GSR sensor is continuously sampled by the Arduino's 10-bit analog-to-digital converter (ADC). The digitized signal is processed using predefined threshold values to determine whether the user's physiological response is normal or abnormal. Under normal conditions, the GSR values remain within a stable range. Sudden or sustained increases in GSR readings indicate abnormal stress or neurological response and are treated as potential warning signs of silent neuroinflammation.

1) GSR Sensor Technical Specifications: Table III presents the technical specifications and pin description of the GSR sensor used in the proposed system.

The GSR sensor plays a crucial role in the proposed system by enabling continuous monitoring of physiological stress responses associated with neurological health. When combined with other biomedical sensors such as ECG, temperature, pulse oximeter, and MEMS sensor, the GSR sensor enhances the system's ability to identify abnormal physiological patterns at an early stage. This multi-sensor approach significantly improves the reliability of silent neuroinflammation detection and supports proactive healthcare intervention.

TABLE III
GSR SENSOR TECHNICAL SPECIFICATIONS

Parameter	Specification
Sensor Type	Galvanic Skin Response Sensor
Operating Voltage	3.3V – 5V
Output Type	Analog Voltage
Measurement Range	10 kΩ – 10 MΩ
Response Time	Less than 1 second
Number of Electrodes	2
Arduino Interface	Analog Pin (A0)
Application	Stress and Nervous System Monitoring

Fig. 4. DS18B20 Temperature Sensor and Pin Configuration

D. Temperature Sensor

The DS18B20 digital temperature sensor is used in the proposed wearable IoT-based health monitoring system to continuously measure the body temperature of the user. Body temperature is a critical physiological parameter, as abnormal temperature variations are often associated with inflammatory responses and physiological imbalance. In cases of silent neuroinflammation, prolonged inflammatory activity may lead to subtle but persistent changes in body temperature, making continuous monitoring essential for early detection.

Fig. 4 shows the DS18B20 temperature sensor and its pin configuration used in the proposed system.

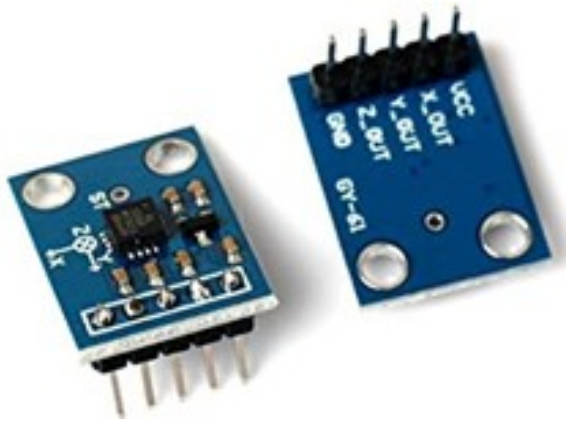

Fig. 5. MEMS Accelerometer Module

The use of a digital temperature sensor enhances the reliability and accuracy of temperature measurement in the proposed system. When combined with other biomedical sensors such as ECG, GSR, pulse oximeter, and MEMS sensor, continuous temperature monitoring strengthens the system's capability to detect early physiological changes associated with silent neuroinflammation.

E. Pulse Oximeter

The pulse oximeter is used in the proposed wearable IoT-based health monitoring system to measure blood oxygen saturation (SpO_2) and pulse rate, which are critical indicators of cardiovascular and respiratory health. Blood oxygen saturation represents the percentage of oxygen-carrying hemoglobin in the blood, and any prolonged reduction in SpO_2 levels may indicate impaired oxygen delivery to vital organs, including the brain. In individuals affected by silent neuroinflammation, inflammatory processes can influence cerebral blood flow and autonomic regulation, potentially leading to abnormal oxygen saturation levels.

F. MEMS Sensor

The MEMS (Micro-Electro-Mechanical Systems) sensor used in the proposed system is a three-axis accelerometer that detects motion, orientation, and posture of the user. MEMS accelerometers measure acceleration forces along the X, Y, and Z axes, enabling the analysis of movement patterns and body position in real time. These sensors are widely used in wearable healthcare devices due to their small size, low power consumption, and high sensitivity.

Fig. 5 shows the MEMS accelerometer module used in the proposed system.

G. ECG Sensor

The Electrocardiogram (ECG) sensor is used in the proposed system to monitor the electrical activity of the heart and extract vital parameters such as heart rate and cardiac rhythm. The heart generates small electrical impulses during each contraction, and these signals are indicative of the functional state of the cardiovascular and autonomic nervous systems. Neurological conditions, including silent neuroinflammation, often influence autonomic nervous system regulation, which can result in abnormal heart rate variability, irregular rhythms, or stress-induced cardiac responses. In the proposed wearable health monitoring system, the ECG sensor continuously captures these electrical signals through surface electrodes placed on the user's body.

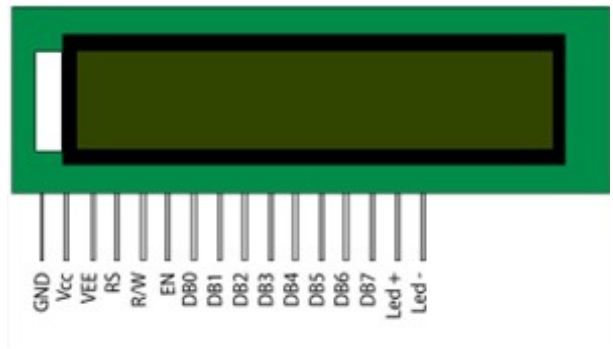

Fig. 6. Pin Configuration of 16×2 LCD Module

H. LCD Display

A 16×2 Liquid Crystal Display (LCD) module is used in the proposed wearable IoT-based health monitoring system to display real-time sensor readings and system status messages. The LCD serves as an effective human–machine interface, enabling users to continuously observe physiological parameters such as heart rate, body temperature, blood oxygen saturation (SpO_2), and alert conditions without requiring any external monitoring device.

The 16×2 LCD consists of two rows with sixteen character positions per row and is controlled by an internal HD44780-compatible LCD controller. The display can operate in either 4-bit or 8-bit data mode. In the proposed system, the LCD is interfaced with the Arduino Uno using control and data registers, allowing reliable communication while minimizing the number of I/O pins required.

In normal operating conditions, the Arduino Uno continuously updates the LCD with stable sensor readings and displays messages such as *"NORMAL STATUS"*. When any sensor value exceeds predefined threshold limits—indicating a potential abnormal physiological condition associated with silent neuroinflammation—the LCD displays warning messages such as *"ABNORMAL CONDITION"* or *"ALERT"*, enabling immediate user awareness.

Fig. 6 shows the pin configuration of the 16×2 LCD module used in the proposed system.

1) LCD Pin Description: Table IV describes the pins of the 16×2 LCD module and their respective functions.

The LCD display significantly enhances the usability of the proposed system by providing continuous visual feedback regarding the user's physiological condition. When combined with the buzzer alert mechanism, the LCD ensures that abnormal conditions are clearly communicated through both visual

TABLE IV
LCD PIN DESCRIPTION

Pin No.	Pin Name	Function
1	GND	Ground
2	VCC	Power supply (5V)
3	VEE	Contrast adjustment
4	RS	Register Select (Command/Data)
5	RW	Read/Write selection
6	EN	Enable signal
7–14	DB0–DB7	Data bus lines
15	LED+	Backlight positive terminal
16	LED	Backlight ground

Fig. 7. Buzzer Alert Module

and audible indications, thereby supporting early detection and timely medical intervention for silent neuroinflammation.

I. Buzzer Alert Unit

The buzzer alert unit is used in the proposed wearable IoT-based health monitoring system to provide an immediate audible warning when abnormal physiological conditions are detected. Audible alerts play a critical role in healthcare monitoring systems, especially in wearable devices, as they ensure that the user or nearby caregivers are instantly notified without the need to continuously observe the display.

In the proposed system, the buzzer is interfaced with the Arduino Uno microcontroller through a digital output pin. The microcontroller controls the buzzer based on the processed sensor data obtained from the ECG sensor, GSR sensor, temperature sensor, pulse oximeter, and MEMS sensor. Under normal operating conditions, when all physiological parameters remain within predefined threshold limits, the buzzer remains in the OFF state.

Fig. 7 shows the buzzer module used in the proposed wearable health monitoring system.

The buzzer alert mechanism enhances the effectiveness of the proposed system by providing real-time emergency notification. When combined with the LCD display, the system ensures that abnormal conditions are communicated through both visual and audible indicators, thereby improving user awareness and supporting timely healthcare response.

V. SOFTWARE IMPLEMENTATION

The software implementation of the proposed wearable IoT-based health monitoring system is developed using the

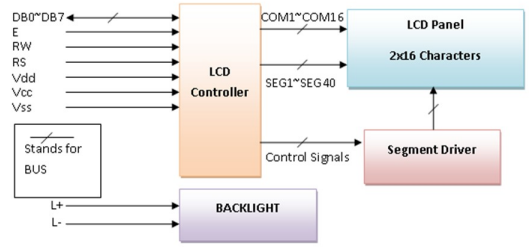

Fig. 8. LCD Controller Architecture Used in Software Implementation

Arduino Integrated Development Environment (IDE) and programmed in Embedded C. The Arduino IDE provides a user-friendly platform for writing, compiling, and uploading programs to the Arduino Uno microcontroller. It also supports a wide range of built-in libraries that simplify sensor interfacing, data acquisition, and display control.

The software is designed to operate in a continuous loop, enabling real-time monitoring of physiological parameters. Initially, the program configures the input and output pins, initializes all biomedical sensors, and sets up the LCD display. After initialization, the Arduino continuously reads data from the connected sensors, including ECG, GSR, temperature sensor, pulse oximeter, and MEMS sensor.

Each sensor value is processed individually and compared with predefined threshold values stored in the program. These threshold values represent normal physiological ranges. If all sensor readings remain within their respective safe limits, the system operates in normal monitoring mode, and the LCD displays real-time sensor values along with a “NORMAL STATUS” message. In this state, the buzzer remains turned OFF.

If any sensor value exceeds its predefined threshold—such as abnormal heart rate, low oxygen saturation, elevated body temperature, abnormal skin conductance, or irregular motion—the software identifies the condition as abnormal. In such cases, the Arduino activates the buzzer alert and updates the LCD display with warning messages such as “ABNORMAL CONDITION” or “ALERT”. This ensures immediate notification to the user and supports timely medical attention.

Fig. 8 illustrates the internal working architecture of the LCD controller used in the software implementation. The controller interprets command and data signals from the Arduino and drives the LCD panel accordingly.

The software structure follows a modular approach, where sensor reading, data processing, display update, and alert generation are handled in separate logical blocks. This improves program readability, simplifies debugging, and allows future enhancements such as cloud connectivity or mobile application integration.

TABLE V
LCD CONTROL AND DISPLAY COMMANDS USED IN SOFTWARE

Command (Hex)	Description
0x01	Clears the entire display screen
0x02	Returns cursor to home position
0x06	Moves cursor to the right after data write
0x0C	Turns display ON and cursor OFF
0x10	Shifts cursor position to the left
0x14	Shifts cursor position to the right
0x28	Sets 4-bit mode with 2-line display
0x38	Sets 8-bit mode with 16×2 display

A. LCD Control and Display Commands

The LCD module is controlled through a set of standard command instructions that manage display behavior, cursor movement, and data writing. These commands are sent from the Arduino to the LCD controller using control pins and data lines. Table V lists the LCD commands used in the software implementation.

The software ensures synchronized operation between sensor data acquisition, display update, and alert generation. By continuously processing real-time physiological data and responding immediately to abnormal conditions, the software plays a vital role in enabling early detection of silent neuroinflammation and improving the overall reliability of the proposed wearable health monitoring system.

VI. ALGORITHM

The algorithm of the proposed wearable IoT-based health monitoring system is designed to continuously acquire physiological data, analyze sensor readings, and generate alerts in case of abnormal conditions associated with silent neuroinflammation. The algorithm operates in a real-time loop to ensure uninterrupted monitoring and timely response.

1) Power ON the system and supply regulated voltage to the Arduino Uno microcontroller and all connected peripheral devices.
2) Initialize the Arduino Uno by configuring the digital and analog input/output pins according to the connected sensors and output devices.
3) Initialize the biomedical sensors, including the ECG sensor, GSR sensor, temperature sensor, pulse oximeter, and MEMS sensor, to ensure stable and accurate data acquisition.
4) Initialize the 16×2 LCD display by sending the required configuration commands to set display mode, cursor behavior, and screen clearing.
5) Define threshold values for each physiological parameter based on normal reference ranges, including heart rate, skin conductance, body temperature, blood oxygen saturation, and motion activity.
6) Continuously read raw sensor data from all connected sensors using the Arduino's analog-to-digital converter and digital input interfaces.
7) Convert the raw sensor data into meaningful physiological parameters such as heart rate (beats per minute), body temperature (in degrees Celsius), SpO_2 percentage, skin conductance levels, and motion values.
8) Compare each calculated physiological parameter with its corresponding predefined threshold values.
9) If all sensor readings are within the normal range, classify the condition as normal and update the LCD display with real-time sensor values along with a "NORMAL STATUS" message.
10) If any sensor reading exceeds its threshold value, classify the condition as abnormal and display warning messages such as "ABNORMAL CONDITION" or "ALERT" on the LCD.
11) Activate the buzzer alert mechanism to provide an audible warning whenever an abnormal physiological condition is detected.
12) Maintain the alert state until the sensor values return to normal operating ranges or the system is manually reset.
13) Repeat the entire monitoring process continuously to enable real-time and long-term health surveillance.

The proposed algorithm ensures synchronized operation between data acquisition, processing, display, and alert generation. By continuously evaluating multiple physiological parameters and responding immediately to abnormal conditions, the algorithm plays a critical role in enabling early detection of silent neuroinflammation and supporting proactive healthcare intervention.

TABLE VI
OBSERVED OUTPUT BEHAVIOR OF THE SYSTEM

Parameter	Normal Condition	Abnormal Condition
Body Temperature	36.5–37.5°C	>37.5°C
Heart Rate	60–100 bpm	<60 or >100 bpm
SpO_2	>95%	<95%
LCD Message	NORMAL STATUS	ALERT / WARNING
LED Indicator	Green ON	Red ON
Buzzer	OFF	ON

VII. CONCLUSION AND FUTURE SCOPE

This paper presented a wearable IoT-based health monitoring system aimed at the early detection of silent neuroinflammation through continuous and non-invasive physiological monitoring. The proposed system integrates multiple biomedical sensors, including ECG, Galvanic Skin Response (GSR), temperature sensor, pulse oximeter, and MEMS motion sensor, to capture real-time physiological and movement-related parameters associated with neurological health.

The Arduino Uno microcontroller serves as the core processing unit, performing threshold-based analysis to classify physiological conditions as normal or abnormal. Real-time sensor readings and system status are displayed on a 16×2 LCD, while an audible buzzer provides immediate alerts during abnormal conditions. This dual alert mechanism enhances user awareness and supports timely medical attention. Experimental results demonstrate that the system can effectively

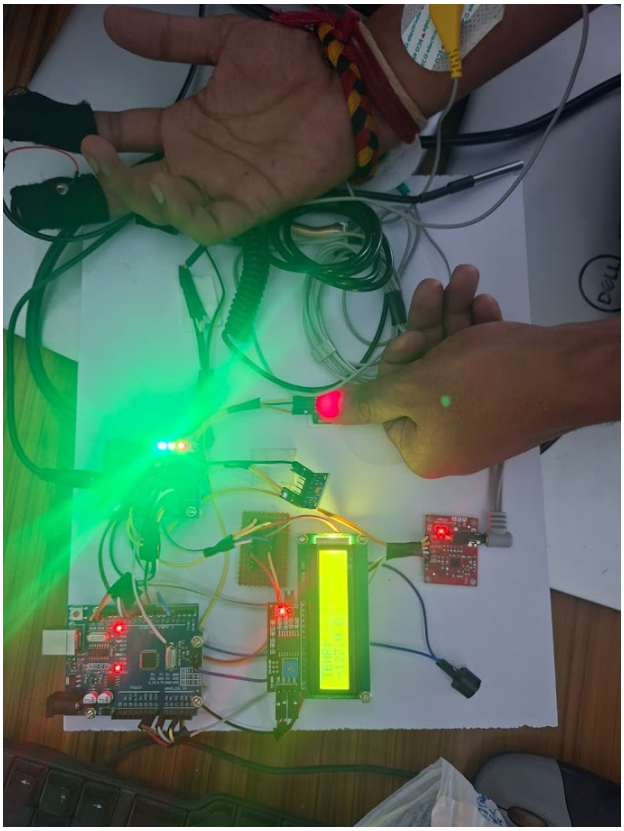

Fig. 9. Hardware output showing normal physiological condition with NORMAL STATUS displayed on the LCD

distinguish between normal and abnormal physiological states, validating the feasibility and reliability of the proposed design.

The system is cost-effective, easy to implement, and suitable for long-term wearable use, making it a promising solution for proactive healthcare monitoring. By enabling early risk identification, the proposed system has the potential to reduce complications associated with delayed diagnosis of silent neuroinflammation.

As future enhancements, the system can be extended with cloud-based data storage to maintain long-term physiological records, enabling doctors and caregivers to remotely monitor patient health and analyze historical trends. Machine learning techniques can be incorporated to perform intelligent health prediction by identifying complex patterns in sensor data, thereby improving detection accuracy and reducing false alerts. Additionally, further miniaturization of hardware components and integration with a mobile health application can enhance user comfort, accessibility, and real-time interaction, making the system more practical for everyday wearable healthcare applications.

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